[SCHOOL SEAL] [DIVISION SEAL] [DepEd SEAL]

Republic of the Philippines

Department of Education

Region V – Division of [PROVINCE]

[SCHOOL NAME IN BOLD CAPS]

[Municipality], [Province]

KALINGA: A LOW-COST BAYMAX-INSPIRED TOUCH-SCREEN HEALTH PENDANT WITH GPS, 4G CALLING, VITAL-SIGN MONITORING, FALL ALERTS, TIMED PILL DISPENSING, AND A FAMILY APP

Submitted to the Scientific Review Committee as an Official Entry to the

2026–2027 National Science and Technology Fair

Science Innovation Expo (SIE) – Team Category

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PATENT / COPYRIGHT APPLICATION

[INSERT HERE]

The IPOPHL application or certificate (Utility Model, Industrial Design, or Copyright) must already have been applied for and secured before the entry is accepted at the Regional Fair (SIE entries only).

ISEF FORMS

[INSERT HERE]

International Rules for Pre-College Science Research forms, signed by the research adviser and the researchers. The forms that apply to this project are listed below.

Form	Title	Applies?
1	Checklist for Adult Sponsor	Yes
1A	Student Checklist	Yes
—	Research Plan / Project Summary	Yes
1B	Approval Form	Yes
3	Risk Assessment Form	Yes
2	Qualified Scientist Form	Only if done in a research institution
4	Human Participants Form	Yes (IRB approval and informed consent)
5A / 5B	Vertebrate Animal Forms	No
6A / 6B	Potentially Hazardous Biological Agents	No
7	Continuation / Research Progression	No

RESEARCH PLAN

A. Rationale

Many older adults and people with chronic illness stay home alone while their families work or live abroad.

Falls are a leading cause of injury and death among adults over 60, and long waits for help make them worse (World Health Organization [WHO], 2021).

About half of patients on long-term medicines do not take them as prescribed (WHO, 2003).

Older adults with memory problems may wander and get lost, and many cannot handle a smartphone to call for help.

Smartwatches, GPS trackers, alert services, and pill dispensers each cover only part of the problem, and together they cost more than most families can pay.

Low-cost touch-screen boards, GPS and 4G modules, 3D printing, and free cloud databases now make it possible to put these jobs into one wearable that the family can watch and call.

B. Research Question or Problem

This study aimed to develop and evaluate KALINGA, a low-cost Baymax-inspired touch-screen health pendant with GPS, 4G calling, vital-sign monitoring, fall alerts, timed pill dispensing, and a family app.

1. How closely do KALINGA's temperature, pulse rate, and SpO_2 readings agree with reference instruments (mean bias, 95% limits of agreement, mean absolute error, and Arms)?
2. How accurately does KALINGA detect simulated falls and ignore daily activities, and how fast do fall, SOS, safe-zone, and inactivity alerts reach the app and by SMS?
3. How accurate is KALINGA's GPS location (error in meters), and how reliably do safe-zone alerts and 4G voice calls work (connection rate and clarity rating)?
4. How accurately and punctually does the pill carousel present each scheduled dose, and how well does it record doses as taken, late, or missed?
5. What share of readings and alerts reach the app, and how long does one battery charge last?

6. How do older adults and family caregivers rate the usability of the pendant and the app?
7. How much does KALINGA cost compared with commercial products that do the same jobs?

C. Goals, Expected Outcomes, and Hypotheses

The project is expected to produce a working pendant and app and to show whether the pendant's readings, alerts, and doses are accurate and timely enough for home monitoring at a lower cost.

- $H_0$1: There is no significant difference between KALINGA's readings and the reference readings for temperature, pulse rate, and SpO_2.
- $H_0$2: KALINGA's fall decision is not associated with the true event type (fall or daily activity).
- $H_0$3: The mean System Usability Scale score of KALINGA is not significantly different from 68, the average score of products in general.

D. Procedures

1. Design the pendant shell in CAD and 3D-print it in PLA.
2. Wire the ESP32-S3 touch board, sensors, GPS, 4G module, speaker, microphone, servo, and battery.
3. Program the touch-screen interface; set up Firebase and the KALINGA Monitor app.
4. Calibrate the thermometer, pulse oximeter, and fall thresholds on a separate calibration set.
5. Run agreement trials with consenting volunteers against reference instruments.
6. Simulate falls (researchers only, on gym mats) and daily activities; time the alerts and SMS.
7. Test GPS at surveyed points, safe-zone exits, and 4G calls in both directions.
8. Run five days of doses, battery tests, and usability sessions with older adults and caregivers.

E. Risk and Safety

- Human participants: IRB approval and informed consent before testing; non-invasive checks only.

- Simulated falls only by the researchers onto 20-cm gym mats, with pads and adult supervision.
- LiPo battery: protected charger, no puncture or heat; charging only under supervision.
- 3D printer and soldering: adult supervision, ventilation, heat-resistant gloves.
- Health data: private Firebase account, coded ids, deleted after the fair (RA 10173).
- Pills in testing: placebo candy tablets only; no real medicines are dispensed.
- 4G module: certified module and antenna; calls and SMS only to consenting family numbers.
- Location data: visible only to the family's accounts; tracking only with the wearer's consent.

F. Data Analysis

- Bland–Altman bias and 95% limits of agreement, MAE, Arms; paired t-test ($H_0$1).
- Confusion matrix, sensitivity, specificity; chi-square test of independence ($H_0$2).
- Percentages for correct slot, on-time doses, delivered messages; mean battery life.
- System Usability Scale; one-sample t-test against 68 ($H_0$3). $\alpha = 0.05$; Excel and JASP.
- GPS error (m) from surveyed points; call connection rate and clarity rating (1–5); safe-zone alert rate.

G. Bibliography

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Photographs, flowcharts, and schematics carry a bold left-aligned label instead of a numbered caption and are not listed here.

ABSTRACT

Many older adults and people with chronic illness spend the day alone while their families work, and a fall, a missed dose, getting lost, or a rising fever may go unnoticed for hours. Smartwatches, GPS trackers, alert services, and pill dispensers each solve one part and together cost more than most families can pay. Inspired by the soft healthcare companion of the animated film Big Hero 6, this study developed KALINGA, a 3D-printed pendant built on an ESP32-S3 board with a 1.28-in round touch screen. It checks forehead temperature, pulse rate, and SpO_2 and asks about pain and warning symptoms; counts steps and active minutes; detects falls and long inactivity; tracks GPS location with a safe zone; makes and answers 4G voice calls and sends SMS alerts with a map link; and turns a four-slot pill carousel to present each dose. Data go through Wi-Fi to Firebase and appear live in the KALINGA Monitor app. Against reference instruments, readings had mean biases of ____ °C, ____ bpm, and ____% SpO_2 (Arms = ____%). Falls were detected with ____% sensitivity and ____% specificity, and alerts arrived in ____ s. GPS error was ____ m, ____% of calls connected, and ____% of doses were presented correctly and on time. The battery lasted ____ h, and users gave a System Usability Scale score of ____ (p = ____ against 68). The pendant cost PHP ____. KALINGA brings monitoring, medicine timing, location, and two-way calls into one affordable wearable.

Keywords: wearable health monitor; GPS; 4G calling; fall detection; pill dispenser; touch screen; elderly care

I. INTRODUCTION

Many older adults and people with chronic illness stay home alone while their families work long hours or abroad (Republic Act No. 9994, 2010).

Falls are a leading cause of injury and death among adults over 60, and lying on the floor for a long time makes the harm worse (WHO, 2021).

About half of patients on long-term medicines miss or mistime their doses (WHO, 2003).

Smartwatches check pulse and detect falls, GPS trackers show location, medical alert pendants call for help, and automatic dispensers time pills, but each does only one part and many need monthly fees.

[NEED: the specific family, barangay, or senior citizens' group where the researchers observed the problem, what was observed, and when. Community Connection and Impact is scored in the Science Innovation Expo.]

There is a need for one low-cost wearable that monitors basic health signs, times medicines, shows location, and lets the older adult and the family call each other with one touch.

In response to the aforementioned problems, the researchers aimed to develop KALINGA, a 3D-printed, Baymax-inspired touch-screen health pendant with vital-sign and symptom checks, step counting, fall, SOS, and safe-zone alerts, 4G calling, a four-slot pill carousel, and the KALINGA Monitor app, and to evaluate its accuracy, timeliness, usability, and cost.

Features and Specification

- Product: KALINGA, a white, rounded, Baymax-inspired pendant on a breakaway cord, with an original two-dot face.
- Size and weight: about 66 × 91 × 43 mm; ____ g with battery and pills.
- Touch screen: 1.28-in round color display with large CHECK, CALL, and SOS buttons and Filipino prompts.
- Health monitoring: forehead temperature (MLX90614), pulse rate and SpO_2 (MAX30102 pad on the back), pain score and warning symptoms (dizziness, shortness of breath, chest pain), steps and active minutes.
- Safety: fall detection with a 10-second cancel window and automatic call to the first contact, SOS button, no-movement alert, GPS safe zone around home.

- Calling: 4G LTE voice calls to up to three family contacts at one touch; family calls are answered automatically after 15 s; SMS alerts carry a Google Maps link.
- Medicine: four-slot pill carousel turned by an SG90 servo; the user taps NAINOM after taking the dose.
- App: KALINGA Monitor shows live status, a map, alerts, vital-sign and step trends, symptoms, the medicine schedule and dose log, contacts, and the safe zone; it can call the pendant.
- Power: 3.7 V 1500 mAh LiPo charged by USB-C; target of one day per charge.
- Intended use: home and community monitoring by families. It screens and alerts; it does not diagnose or treat.

Market Trends and Opportunities

The number of Filipino families with an older member at home grows each year, and more family members work away from home. Smartwatches, GPS trackers, medical alert pendants, and pill dispensers are sold online, but a family needs several of them to cover what KALINGA does. Table 1 compares their costs. Prices are to be filled in from the researchers' own canvass of local stores.

Table 1

Cost Comparative Analysis of KALINGA and Commercial Products

Product	Price (PHP)	Vitals	Fall/SOS	GPS	Calls	Timed dose
KALINGA pendant + app	4,770 (est.)	Yes	Yes	Yes	Yes	Yes
Smartwatch with SpO_2, fall detection, LTE	____	Yes	Some	Yes	Some	No
GPS tracker for seniors	____ + load	No	SOS	Yes	Yes	No
Medical alert pendant with service	____ + ____/month	No	Yes	Some	Yes	No
Automatic pill dispenser	____	No	No	No	No	Yes

Note. Table created by the researchers, 2026.

Scientific Basis

Pulse oximeters shine red and infrared light through a fingertip; the ratio of light absorbed with each heartbeat gives SpO_2, and the beat interval gives pulse rate (Allen, 2007; Jubran, 2015). ISO 80601-2-61 accepts an SpO_2 Arms of 4% or less (International Organization for Standardization [ISO], 2017b).

Every warm surface gives off infrared energy that rises with its temperature (Fraden, 2016). The MLX90614 measures this energy from the forehead; because skin is cooler than the body core, a calibration equation is applied, as in clinical forehead thermometers tested under ISO 80601-2-56 (ISO, 2017a).

A fall shows as a short drop in acceleration (free fall), a sharp peak (impact), and then stillness. Thresholds on these three phases detect falls with high sensitivity and specificity (Bourke et al., 2007).

A GPS receiver finds its position from the timing of signals from several satellites; under open sky, phone-grade receivers are usually accurate to about 5 m (GPS.gov, n.d.). A safe zone is a circle around home; the distance from home is computed from the two coordinates.

Pain scores and warning symptoms such as chest pain or shortness of breath add what sensors cannot measure, and are flagged for prompt follow-up.

Timed reminders and pre-sorted doses are among the simplest ways to improve medicine adherence (WHO, 2003). A new measuring device must be compared with a reference using limits of agreement, not correlation alone (Bland & Altman, 1986), and usability can be measured with the 10-item System Usability Scale, whose average across products is 68 (Brooke, 1996; Sauro, 2011).

Statement of the Problem

This study aimed to develop and evaluate KALINGA. Specifically, it sought to answer the following questions:

1. How closely do KALINGA's temperature, pulse rate, and SpO_2 readings agree with reference instruments (mean bias, 95% limits of agreement, mean absolute error, and Arms)?
2. How accurately does KALINGA detect simulated falls and ignore daily activities, and how fast do fall, SOS, safe-zone, and inactivity alerts reach the app and by SMS?
3. How accurate is KALINGA's GPS location (error in meters), and how reliably do safe-zone alerts and 4G voice calls work (connection rate and clarity rating)?
4. How accurately and punctually does the pill carousel present each scheduled dose, and how well does it record doses as taken, late, or missed?
5. What share of readings and alerts reach the app, and how long does one battery charge last?

6. How do older adults and family caregivers rate the usability of the pendant and the app?
7. How much does KALINGA cost compared with commercial products that do the same jobs?

Hypotheses

The following null hypotheses were tested at the 0.05 level of significance:

- $H_0$1: There is no significant difference between KALINGA's readings and the reference readings for temperature, pulse rate, and SpO_2.
- $H_0$2: KALINGA's fall decision is not associated with the true event type (fall or daily activity).
- $H_0$3: The mean System Usability Scale score of KALINGA is not significantly different from 68, the average score of products in general.

II. MATERIALS AND METHODS

Procurement of the Materials

Components were bought from local electronics shops and online stores; the shell was printed on a ____ 3D printer [NEED: printer model, or the print shop used]. A prepaid nano-SIM with load was used for calls and SMS. The reference instruments were borrowed from the school clinic. Expenses are in Appendix A.3.

Table 2

Materials and Components of KALINGA

Material / Component	Quantity	Purpose
ESP32-S3 1.28-in round touch LCD board	1	Controller, touch screen, motion sensor, charger
MLX90614 infrared thermometer	1	Forehead temperature
MAX30102 pulse oximeter module	1	Pulse rate and SpO_2
ATGM336H GPS module	1	Location and safe zone
A7670 4G LTE module, SIM, speaker, microphone	1 set	Voice calls and SMS
SG90 micro servo and printed carousel	1	Pill dispensing
Vibration motor	1	Reminders and alarms
LiPo 1500 mAh	1	Power
White PLA, black paint	75 g	3D-printed shell
Reference thermometer and pulse oximeter	1 each	Reference readings (borrowed)
20-cm gym mats, knee and elbow pads	2; 1 set	Simulated falls

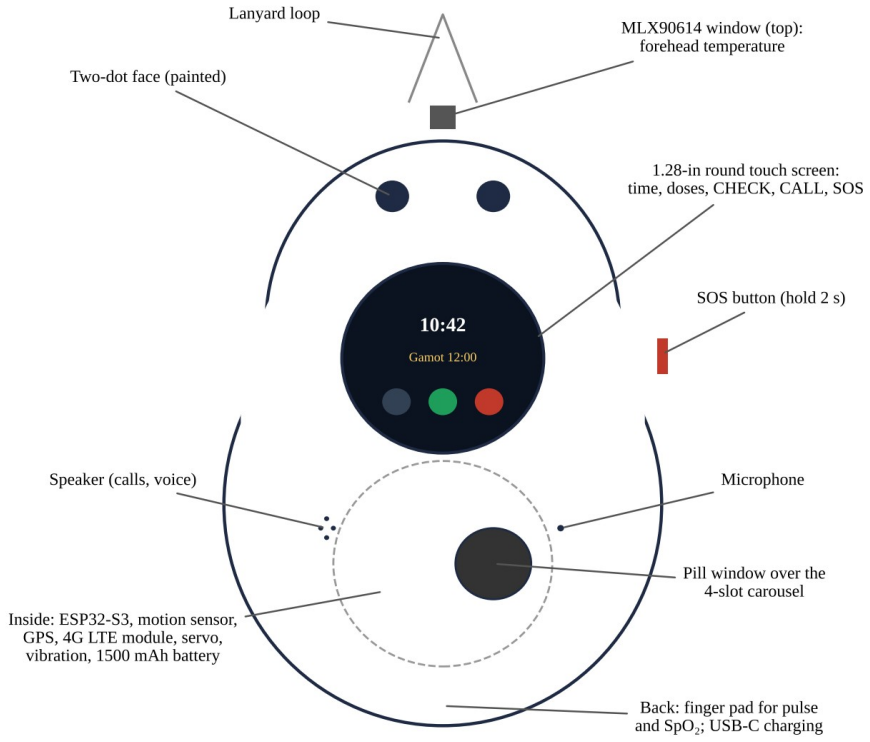
Note. Table created by the researchers, 2026.

Construction of the Product

Design and 3D Printing of the Shell

The shell was modeled as a front shell, a back shell, and a pill carousel with a parametric script that outputs STL files (Appendix, Dimensions of the Product). The front has a 33-mm round window for the touch screen, a top window for the thermometer, a pill window, a speaker grille, a microphone hole, and an SOS button; the back has the finger pad window. Parts were printed in white PLA with a 0.4-mm nozzle, 0.2-mm layers, 3 walls, and 20% infill, with tree supports; the eyes were painted black and the edges sanded.

Design of the KALINGA Pendant



Front view, actual size about 66 × 91 × 43 mm, 3D-printed in white PLA.

Note. Image created by the researchers, 2026.

3D-Printed Parts: Back Shell, Pill Carousel, and Front Shell

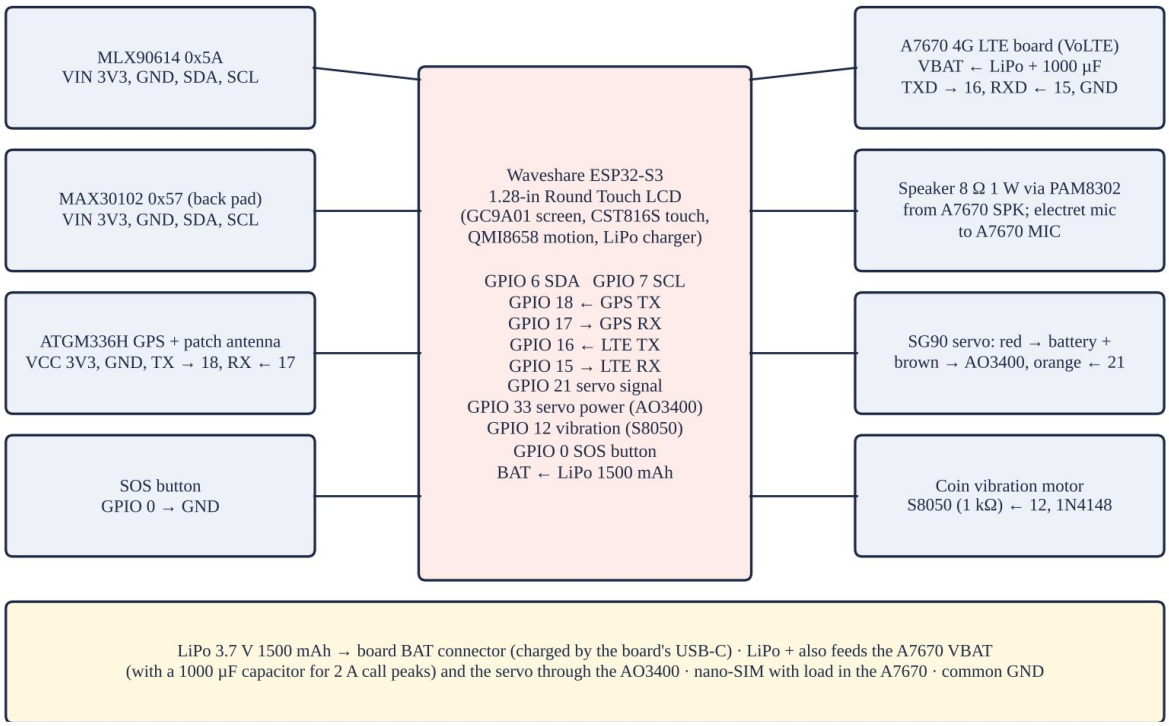


Note. Image created by the researchers, 2026.

Electronics

The Waveshare ESP32-S3 1.28-in round touch board carries the screen, the touch controller, a QMI8658 motion sensor, and a LiPo charger. The MLX90614 and MAX30102 share its I²C bus at 100 kHz. An ATGM336H GPS module and a SIMCom A7670 4G LTE module connect to two serial ports; the A7670 drives a small speaker through a PAM8302 amplifier and takes an electret microphone, and a 1000 µF capacitor supports its current peaks during calls. Connections are in Appendix A.4.

Wiring Diagram of KALINGA



Note. Image created by the researchers, 2026.

Pill Carousel

The 38-mm carousel has four pockets at 45°, 90°, 135°, and 180° and a solid sector at 0°, which closes the window. At each dose time the servo turns the right pocket under the window; the pendant vibrates and shows the medicine every 2 minutes for up to 30 minutes until the user taps NAINOM. Doses are recorded as taken (within 10 minutes), late, or missed, and a missed dose alerts the family.

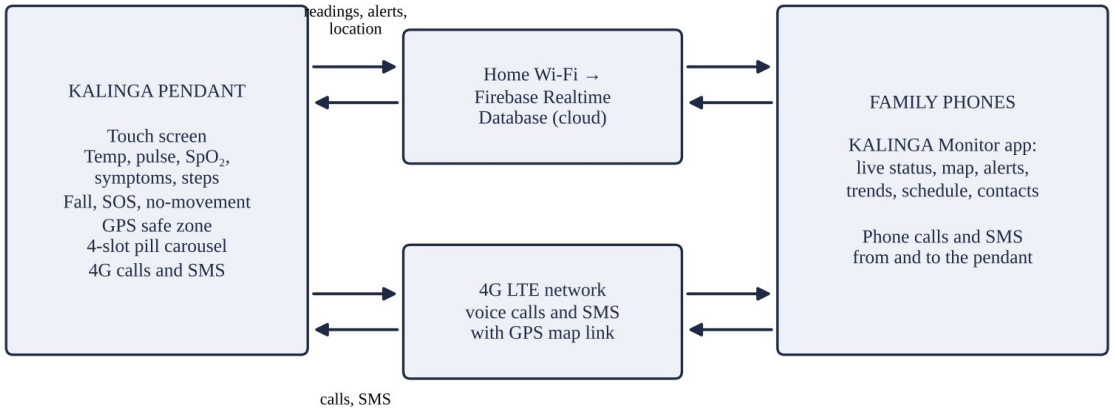
Touch-Screen Interface, Calls, and Location

The home screen shows the time, the next dose, battery, and steps, with CHECK, CALL, and SOS buttons. CHECK runs the temperature and pulse checks and then asks for a pain score with + and − buttons and three yes/no questions. CALL lists the family contacts set in the app and dials through the A7670; incoming family calls show ANSWER and DECLINE and are answered automatically after 15 s. GPS location is saved every 5 minutes, and leaving the safe zone, a fall, an SOS, or a RED check sends both an app alert and an SMS with a map link. After a fall that is not cancelled, the pendant also calls the first contact.

Program and App

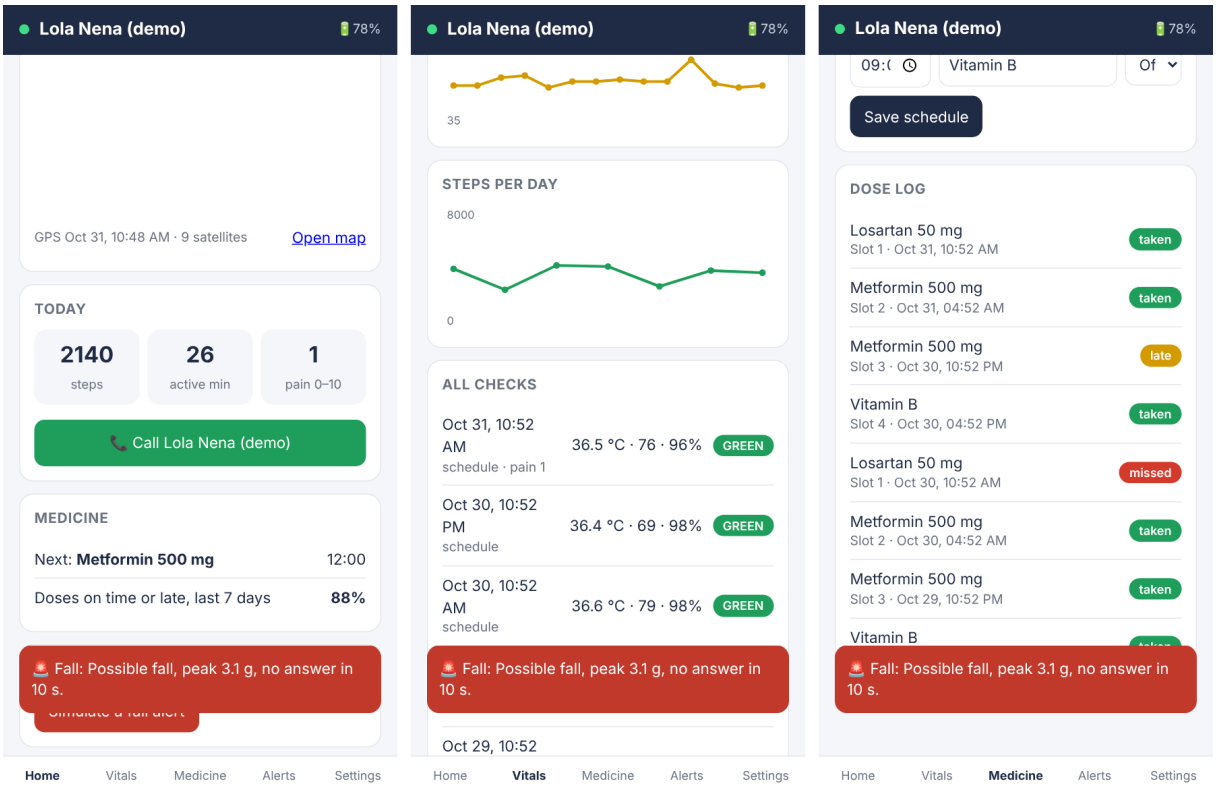
The ESP32-S3 checks for falls and counts steps continuously and, once a minute, sends its status, steps, and location and reads the schedule, contacts, safe zone, and commands from a Firebase Realtime Database through its REST interface (Google, n.d.). The KALINGA Monitor app signs in with Firebase Authentication, receives changes as a live stream, shows the location on a map, and can call the pendant's number. Database rules let only the family's accounts read or write their patient's data.

How Data Flows Between the Pendant and the App



Note. Image created by the researchers, 2026.

KALINGA Monitor App (demo mode): Home, Vitals, and Medicine Screens



Note. Image created by the researchers, 2026.

Data Gathering

Tests were done indoors at 26 to 30 °C. Volunteers rested seated for 5 minutes before each check, and the pendant and the reference instrument were used within 1 minute of each other in alternating order. The first 10 volunteers formed the calibration set and were not used in the agreement analysis. Simulated falls (forward, backward, sideways, and from a chair) were done only by the researchers onto 20-cm gym mats; daily activities were sitting, lying on a bed, bending, walking, and climbing a step. Placebo candy tablets were used in all dose tests. GPS was tested at ten points whose coordinates were read from satellite imagery, with 10 fixes each; the safe zone was tested by walking out of it 10 times. Calls were made 20 times in each direction, and the listener rated clarity from 1 (cannot understand) to 5 (clear).

Table 3

Trials Conducted in the Study

Trial set	What is measured	Number	Reference
Bench temperature	IR reading of a 35.0–40.0 °C target	6 × 10	Probe thermometer
Calibration set	Temperature, PR, SpO_2	10 volunteers × 3	Clinic instruments

Validation set	Temperature, PR, SpO_2	____ volunteers × 3	Clinic instruments
Falls and activities	Fall decision, alert time	____ falls; ____ activities	Video
Doses	Slot, timing, record	5 days × 4 slots	Observer log
GPS and safe zone	Location error; zone alerts	10 points × 10; 10 exits	Satellite imagery
Calls	Connection; clarity 1–5	20 each direction	Listener rating
Delivery and battery	Messages received; hours per charge	5 days; 3 charges	App log
Usability	SUS	____ older adults; ____ caregivers	Questionnaire

Note. Table created by the researchers, 2026.

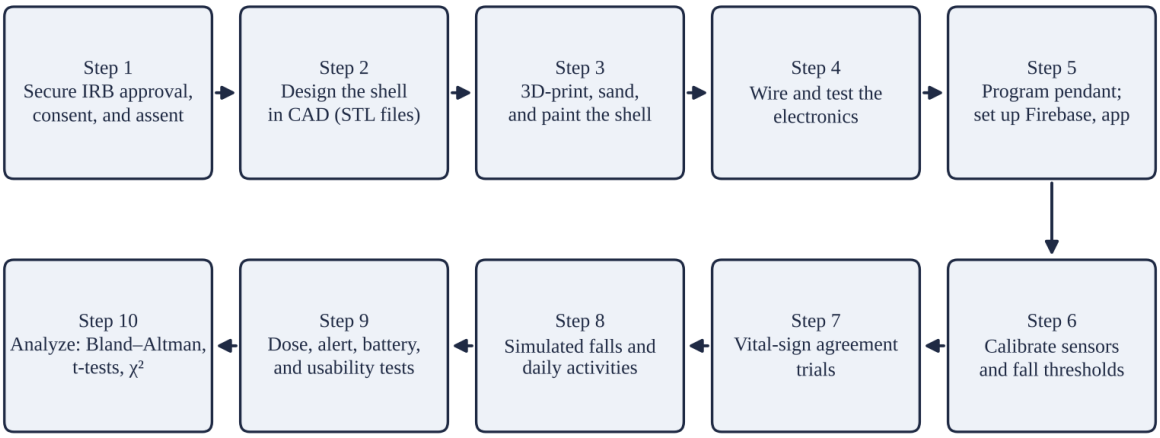
The following formulas were used:

$d = KALINGA - reference$; $Bias = \Sigma d \div n$	where difference and mean bias	(1)
$LoA = Bias \pm 1.96 \times SD(d)$; $Arms = \sqrt{(\Sigma d^2 \div n)}$	where limits of agreement; SpO_2 RMS error	(2)
$Sensitivity = TP \div (TP + FN)$; $Specificity = TN \div (TN + FP)$	where falls detected; activities ignored	(3)
$t_alert = t_app - t_event$	where alert time (s)	(4)
Correct-slot rate = correct slots ÷ doses × 100	where carousel accuracy (%)	(5)
$SUS = 2.5 \times [\Sigma(\text{odd item} - 1) + \Sigma(5 - \text{even item})]$	where System Usability Scale score (0–100)	(6)
GPS error = distance(fix, reference point)	where location error (m)	(7)

$H_0 1$ was tested with a paired t-test for each measure, $H_0 2$ with a chi-square test of independence, and $H_0 3$ with a one-sample t-test against 68, at $\alpha = 0.05$, using Microsoft Excel and JASP.

Flowchart of the Procedure

Steps of the Study



Note. Image created by the researchers, 2026.

III. RESULTS AND DISCUSSION

[NEED: every ____ in this chapter must be replaced with values the researchers measured. No value may be entered that was not measured.]

Agreement of the Vital-Sign Checks

Table 4.1 shows the infrared thermometer against a water-bath target, and Table 4.2 the agreement with the reference instruments in ____ validation volunteers. Temperature had a bias of ____ °C (LoA ____ to ____), pulse rate ____ bpm, and SpO_2 ____% with an Arms of ____%, which ____ the 4% limit. The paired t-test gave p = ____, ____, and ____, so $H_0 1$ is ____.

Table 4.1

Bench Test of the Infrared Thermometer Against a Water-Bath Target

Target (°C)	Probe reading (°C)	ALAGA mean ± SD (°C)	Error (°C)
35.0	____	____ ± ____	____
36.0	____	____ ± ____	____
37.0	____	____ ± ____	____
37.5	____	____ ± ____	____
38.0	____	____ ± ____	____
40.0	____	____ ± ____	____

Note. Table created by the researchers, 2026.

Table 4.2

Agreement of KALINGA With the Reference Instruments

Measure	n	Bias	95% LoA	MAE	Arms	p
Temperature (°C)	____	____	____ to ____	____	____	____
Pulse rate (bpm)	____	____	____ to ____	____	____	____
SpO_2 (%)	____	____	____ to ____	____	____	____

Note. Table created by the researchers, 2026.

Figure 1

Bland–Altman Plots of Temperature, Pulse Rate, and SpO_2

[NEED: Bland–Altman plots from the measured validation data: difference (ALAGA − reference) against the mean of the two, with lines for the bias and the 95% limits of agreement.]

Note. Figure created by the researchers, 2026.

Fall Detection and Alerts

KALINGA detected ____ of ____ simulated falls (sensitivity ____%) and ignored ____ of ____ daily activities (specificity ____%); χ^2 = ____, p = ____, so $H_0$2 is ____ (Table 5.1). Alerts reached the app in a mean of ____ s (Table 5.2).

Table 5.1

Confusion Matrix of KALINGA Fall Decisions

	True fall	Daily activity	Total
KALINGA: fall	TP = ____	FP = ____	____
KALINGA: no fall	FN = ____	TN = ____	____
Total	____	____	____

Note. Table created by the researchers, 2026.

Table 5.2

Time for Alerts to Reach the App and by SMS

Alert	n	App, mean ± SD (s)	SMS, mean ± SD (s)
Fall	____	____ ± ____	____ ± ____
SOS	____	____ ± ____	____ ± ____
Left safe zone	____	____ ± ____	____ ± ____
No movement	____	____ ± ____	____ ± ____
Abnormal check	____	____ ± ____	____ ± ____

Note. Table created by the researchers, 2026.

GPS Location and Calls

The mean GPS error was ____ m (SD ____) outdoors and ____ m indoors near a window, and ____ of 10 safe-zone exits sent an alert (Table 6.1). ____ of 40 calls connected, with a mean clarity rating of ____ (Table 6.2).

Table 6.1

GPS Location Error and Safe-Zone Alerts

Condition	Fixes	Error, mean ± SD (m)	Largest (m)
Open sky	____	____ ± ____	____
Near a window indoors	____	____ ± ____	____
Under a roof outdoors	____	____ ± ____	____
Safe-zone exits alerted	____ / 10	—	—

Note. Table created by the researchers, 2026.

Table 6.2

4G Voice Calls

Direction	Attempts	Connected	Clarity, mean (1–5)
Pendant to family	20	____	____
Family to pendant (auto-answer)	20	____	____

After a fall (automatic)	____	____	____

Note. Table created by the researchers, 2026.

Pill Carousel and Dose Records

Over ____ doses, the carousel presented the correct slot ____% of the time, ____% within 1 minute of schedule, and the dose log matched the observers' record in ____% of doses (Table 7.1).

Table 7.1

Dose Presentation and Records Over Five Days

Measure	Count	Result (%)
Correct slot presented	____ / ____	____
Presented within 1 min of schedule	____ / ____	____
Records matching the observer (taken, late, missed)	____ / ____	____
Missed-dose alerts received	____ / ____	____

Note. Table created by the researchers, 2026.

Data Delivery and Battery Life

____% of readings, ____% of app alerts, and ____% of SMS alerts were received, and one charge lasted ____ h (Table 8.1).

Table 8.1

Data Delivery and Battery Life

Measure	Result
Readings received by the app	____ / ____ (____%)
Alerts received by the app	____ / ____ (____%)
SMS alerts received	____ / ____ (____%)
Battery life per charge, mean ± SD (h)	____ ± ____
Charging time (h)	____

Note. Table created by the researchers, 2026.

Usability

Older adults gave a mean SUS score of ____ and caregivers ____; against 68, t = ____, p = ____, so $H_0$3 is ____ (Table 9.1).

Table 9.1

System Usability Scale Scores

Group	n	Mean ± SD	t (vs 68)	p
Older adults	____	____ ± ____	____	____
Family caregivers	____	____ ± ____	____	____
All users	____	____ ± ____	____	____

Note. Table created by the researchers, 2026.

Discussion

[NEED: explain the results against the literature: why the SpO_2 and temperature biases are as found, which falls were missed and why, where GPS was weakest, what affected call quality, what caused late or missed doses, and what users liked or found hard on the touch screen. Compare cost with Table 1.]

IV. CONCLUSION

Summary of Findings

1. Readings had biases of ____ °C, ____ bpm, and ____% SpO_2, with an SpO_2 Arms of ____%; $H_0$1 is ____.
2. Falls were detected with ____% sensitivity and ____% specificity; alerts arrived in ____ s; $H_0$2 is ____.
3. GPS error was ____ m; ____% of safe-zone exits were alerted; ____% of calls connected (clarity ____).
4. The carousel presented the correct slot in ____% of doses, ____% on time.
5. ____% of readings and alerts reached the app; the battery lasted ____ h.
6. Users gave a SUS score of ____; $H_0$3 is ____.
7. KALINGA cost PHP ____, against PHP ____ for the commercial products in Table 1.

Conclusion

[NEED: one paragraph based on the measured results.] KALINGA joins health monitoring, timed doses, location, two-way calls, and family alerts in one low-cost wearable.

Recommendations

1. Shrink the pendant with a custom circuit board and a smaller dispensing mechanism.
2. Add phone push notifications, indoor location, and a longer-lasting battery.
3. Test with real falls data and with people of different skin tones and ages.
4. File the shell as an Industrial Design and the carousel as a Utility Model with IPOPHL, keeping the design original rather than copying the film character.

V. ACKNOWLEDGEMENT

[NEED: thank the people and offices that helped: adviser, school clinic, volunteers, family, and any partner who printed the shell.]

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VII. APPENDICES

Appendix A. Raw Data, Expenses, and Connections

Table A.1

Raw Data Sheet for Vital-Sign Checks

Code	Trial	ALAGA T (°C)	Ref T (°C)	ALAGA PR	Ref PR	ALAGA SpO_2	Ref SpO_2	Pain	Time (s)
V01	1								
V01	2								
V01	3								
V02	1								
V02	2								
V02	3								

Note. Table created by the researchers, 2026.

Table A.2

Raw Data Sheet for Falls and Daily Activities

Trial	Event	Detected (Y/N)	Cancelled	Peak g	Alert time (s)	Notes
1	Forward fall					
2	Backward fall					
3	Sideways fall					
4	Fall from chair					
5	Sit down					
6	Lie on bed					
7	Bend					
8	Walk					
9	Climb a step					

Note. Table created by the researchers, 2026.

Table A.3

Expense Summary of the Study

Item	Qty	Estimated price (PHP)
Waveshare ESP32-S3 1.28-in round touch LCD board	1	1,300
MLX90614 (GY-906) module	1	550
MAX30102 module	1	180
ATGM336H GPS module with patch antenna	1	300
SIMCom A7670 4G LTE board (VoLTE) with antenna	1	1,200
Prepaid nano-SIM	1	50
PAM8302 amplifier, 8 Ω 1 W speaker, electret microphone	1 set	200
SG90 micro servo	1	120

LiPo 3.7 V 1500 mAh	1	280
Coin vibration motor, AO3400, S8050, 1000 µF, resistors	1 set	80
SOS button, slide switch, perfboard, wires	1 lot	110
PLA filament or print service (about 75 g)	1 lot	300
Black acrylic paint, sandpaper	1 lot	40
Breakaway lanyard, M2 screws	1 set	60
Total (estimated)		**4,770**

Note. Table created by the researchers, 2026.

Table A.4

Pin Connections of KALINGA

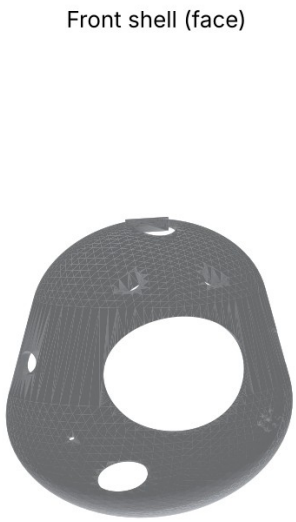
Part	Pin	ESP32-S3 board	Note
MLX90614, MAX30102 (and on-board touch, QMI8658)	SDA / SCL	GPIO 6 / 7	I²C at 100 kHz
ATGM336H GPS	TX / RX	GPIO 18 / 17	UART1, 9600 baud
A7670 4G LTE	TXD / RXD	GPIO 16 / 15	UART2, 115200 baud; VBAT from LiPo + 1000 µF
SG90 servo	signal / power	GPIO 21 / 33	AO3400 switches the servo ground
Vibration motor	S8050 base	GPIO 12	1 kΩ; 1N4148 across motor
SOS button	to GND	GPIO 0	Also the BOOT button
LiPo 1500 mAh	+ / –	BAT connector	Charged by the board's USB-C

Note. Table created by the researchers, 2026.

Dimensions of the Product

Overall size about 66 mm wide, 91 mm tall, and 43 mm deep, including the lanyard loop; round screen window 33 mm; pill carousel 38 mm across with four 9.6-mm pockets. Print files: cad/front_shell.stl, cad/back_shell.stl, and cad/pill_carousel.stl, made by cad/kalinga_pendant.py.

Front Shell, Rendered From the STL File



Note. Image created by the researchers, 2026.

[NEED: photographs of the finished pendant with dimension callouts and its weight.]

Program and App

Pendant program: firmware/kalinga_pendant/kalinga_pendant.ino (ESP32S3 Dev Module). App and website: app/ and index.html; database rules in app/database.rules.json. The engineering notebook gives every build and setup step.

Photographs of the Procedure

[NEED: dated photographs of each step in the flowchart, each with a bold label and the note “Photographs taken by the researchers, 2026.” No faces of volunteers without written permission.]

Informed Consent

Summary read to each participant: “You are invited to try KALINGA, a pendant that checks temperature, pulse, and oxygen and reminds you about medicine. You will sit while it reads your forehead from a short distance and you place one finger on its pad; the clinic's instruments will be used too. You will answer a short questionnaire. It takes about 20 minutes and does not hurt. You will not be asked to fall or take any medicine. Your name will not be recorded.”

[NEED: signed consent and assent forms and the IRB approval.]

Curriculum Vitae

[NEED: one-page curriculum vitae of each researcher with a 2 × 2 photo.]